



THÈSE DE DOCTORAT
DE L'UNIVERSITÉ PSL

Préparée à MINES Paris

**Advanced Hemodynamic Modelling and Simulation of Flow
Diverter in the Treatment of Intracranial Aneurysms**

Soutenue par

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Le Date de soutenance

École doctorale n°364

**Sciences Fondamentales et
Appliquées**

Spécialité

**Mathématiques Numériques,
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Dedication

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Abstract

In the previous chapters, we have developed the numerical framework for VMS in fluid simulations, examined the specifics of brain arterial hemodynamics, and created a tool for modeling Flow-Diverter Stents (FDS) based on geometric constraints. Leveraging the experience and tools acquired, we now apply this framework to three patient cases featuring two variations of intracranial aneurysms, each presenting unique challenges to clinicians. These cases highlight situations where FDS treatment is considered highly promising, despite the still high fatality rates. First, we address the giant aneurysm (> 25 mm) previously discussed. This case will be thoroughly analyzed from medical and anatomical perspectives before delving into the simulation of pre- and postoperative hemodynamics. The second case involves blister-like aneurysms, formations that are barely visible on CT imaging and are characterized by critically thin and fragile walls. Both types of aneurysms are poorly understood and necessitate further research to enhance treatment outcomes. Through our simulations, we demonstrate the capabilities of the framework and, more importantly, provide hemodynamic results for these scarcely documented aneurysms, correlating our findings with existing literature.

Resumé

Dans les chapitres précédents, un framework numérique pour les simulations hémodynamiques utilisant la méthode de VMS a été établi, ainsi que l'adaptation des simulations aux spécificités anatomiques du réseau artériel et l'intégration des stents à diversion de flux, en validant ces approches par des méthodes numériques et des comparaisons avec des données cliniques. À partir de cette base, le framework est appliqué à trois cas cliniques impliquant deux types d'anévrismes intracrâniens, chacun posant des défis spécifiques. Ces cas illustrent des situations où l'utilisation des stents à diversion de flux est particulièrement prometteuse, bien que les taux de mortalité restent élevés. Le premier cas concerne un anévrisme géant (≥ 25 mm), qui sera examiné en détail tant du point de vue médical qu'anatomique avant d'effectuer des simulations des hémodynamiques pré- et postopératoires. Le second cas implique des anévrismes "blister-like", difficilement visibles en imagerie CT et caractérisés par des parois extrêmement fines et fragiles. Ces types d'anévrismes sont encore mal compris et nécessitent une recherche supplémentaire pour améliorer les résultats des traitements. Les simulations permettront de démontrer les capacités du framework et de fournir des données hémodynamiques pour ces anévrismes peu documentés, en les mettant au regard des données et conclusions établies dans la littérature scientifique.

1.1 Delayed Rupture of a Giant Aneurysm

Giant aneurysms present several added challenges compared to their smaller counterparts. Due to their size, they are often symptomatic, causing cerebral palsies, headaches, and other neurological disturbances. Therefore, treating them is often necessary, though such operations are more complex and prone to complications [1]. Most giant aneurysms involve a significant portion of the parent vessel and lack a well-defined neck. As a result, surgical clipping is often unviable. Even deploying coils is rarely feasible since they require sufficient anchorage to remain firmly in place. Moreover, coils face an increased risk of recanalization due to material fatigue and compaction [2]. Stent-assisted coiling has also not been found to improve the likelihood of a positive outcome [1, 3].

Flow-diverter stents, although sometimes difficult to deploy in giant aneurysms, circumvent these issues. Despite their overall good performance [4], giant aneurysms still pose a significant risk of delayed rupture [1, 4, 5]. Since the first observed cases in the 2010s, research teams have proposed several contributing factors and documented their observations. In the following section, we will conduct a detailed analysis of a patient case who unfortunately experienced this dramatic outcome.

1.1.1 Patient Medical Records

The patient was a woman born in January 1963, with no history of smoking, hypertension, or family history of intracranial aneurysms. She experienced vertigo, anxiety, and a strange sensation, prompting her to consult a neurologist. Following an initial consultation, she underwent an MRI scan of the brain, which revealed a giant intracranial aneurysm compressing the left temporal lobe¹. Her neurologist referred her to a neuro-interventionalist for further evaluation. A cerebral arteriography confirmed the presence of a giant aneurysm in the left supra-clinoid ICA, between the ophthalmic and communicating segments. The aneurysm's involvement of the parent artery caused delayed opacification of the left middle cerebral artery. The small anterior communicating artery was visible during compression of the left ICA.

Given the symptoms and risk of rupture, endovascular treatment was recommended and accepted by the patient. One week before the procedure, she began taking aspirin (75 mg/day) and prasugrel² (10 mg/day) to reduce the risk of blood clots. The procedure commenced with a balloon test occlusion, where a balloon was temporarily inflated in the artery to block blood flow. A guiding catheter was placed in the

¹Temporal lobes: Regions of the brain situated on both halves above the ears.

²Prasugrel: A platelet aggregation inhibitor medication.

right and left internal carotid arteries. A Scepter XC 4–11 balloon was positioned in the left ICA upstream of the aneurysm and inflated to stop blood flow. Contrast injection via the right catheter showed a delay of over one second in the opacification of the left cerebral veins compared to the right, indicating that the left carotid artery could not be safely occluded. Therefore, preserving and reconstructing the left ICA with a flow-diverting stent was necessary. A 3 mm FRED Jr flow-diverter was deployed from the left MCA (M1 segment) to the left ICA upstream of the aneurysm, immediately causing contrast stasis in the aneurysm sac. Upon waking up, the patient showed no neurological deficits and continued antiplatelet therapy. At her follow-up 1.5 months later, her clinical exam returned normal results. Six months after the procedure, a follow-up cerebral arteriography showed a patent stent³ with unchanged opacification of the aneurysm sac. Continued dual antiplatelet therapy was advised, with a repeat arteriography scheduled for ten months post-procedure.

Eight months after the intervention, the patient reported a sudden headache. A CT scan revealed spontaneous hyperdensity in the aneurysm, suggesting thrombosis. A week later, she suffered a sudden coma, with a Glasgow Coma Scale score of 4, indicating severe brain injury. A CT scan revealed subarachnoid hemorrhage, intraventricular hemorrhage, and a left temporal hematoma⁴. Cerebral arteriography showed persistent aneurysm opacification, a patent stent, and active contrast leakage from the posteroinferior part of the aneurysm sac. A microcatheter was placed in the aneurysm sac through the stent mesh, and the sac was occluded with coils, halting the bleeding and completely occluding the aneurysm. Sadly, the patient passed away five days after this second intervention.

1.1.2 Arterial Anatomy

Circle of Willis

An MRA scan confirmed an incomplete Circle of Willis (CoW), missing the posterior communicating artery ipsilateral to the aneurysm. The contralateral posterior communicating artery, which appears to have a fetal configuration, supplies most of the blood to the left posterior circulation. The anterior communicating artery, although present, is small, as confirmed during a clamp test. Based on these features, the CoW structure is categorized as a type PCP, a variant found in approximately 7% of the population [6]. Although this CoW configuration carries a 2.3-fold increased risk of aneurysm formation, its effect on rupture risk remains unclear.

The aneurysm is located on the communicating (C7) segment of the supra-clinoidal

³Patent (stent): Maintaining the openness and integrity of the vessel or structure where it has been placed.

⁴Hematoma: A solid swelling of clotted blood within the tissues.

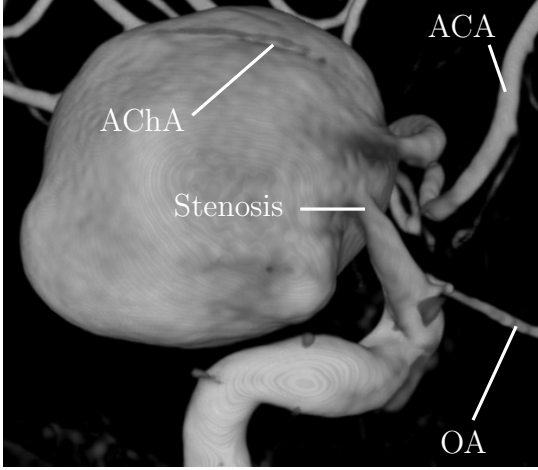


Figure 1.1: Annotated DSA image of the preoperative IA.

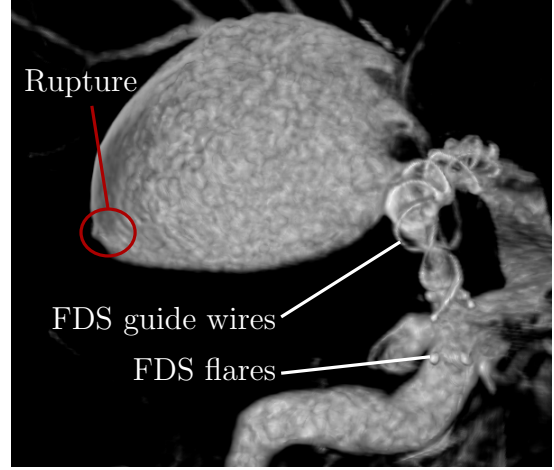


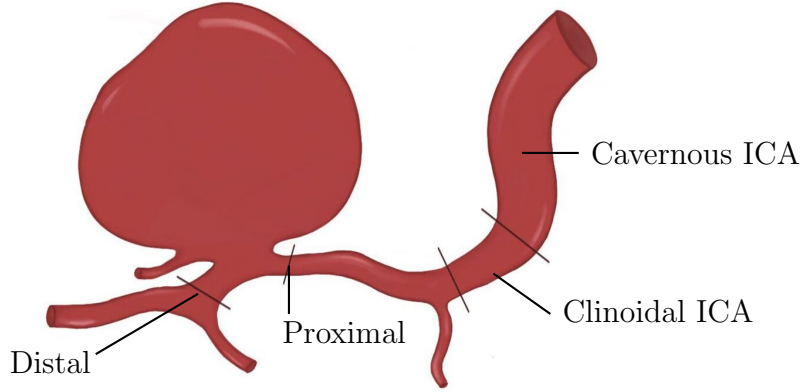
Figure 1.2: Annotated DSA image of the postoperative IA.

ICA, 4,mm before its terminal bifurcation. The anterior choroidal artery (AChA) emanates from the aneurysm, though its flow is barely noticeable on DSA scans, as shown on the left of Fig. 1.1.

Tapering

A crucial factor in this case is the presence of tapering (T) and vessel deformation around the aneurysm. The cavernous ICA has a diameter of 4.16,mm, consistent with reported values [7]. It experiences continuous tapering over the ophthalmic (C6) and communicating (C7) segments, leading to a nearly elliptical vessel at the proximal aneurysm neck. This atypical cross-section translates into a 32% tapering relative to the diameter of the cavernous ICA (see Fig. 1.3). Beyond the aneurysm, the terminal section narrows to 2.15,mm, indicating a 48% stenosis. This value falls outside the standard deviation of the recorded cases by Baz *et al.* [7].

After stent deployment, the proximal vessel sections remain largely unchanged. However, the clinoidal ICA shrinks, with diameters of 3.33,mm, representing a 20% tapering relative to the cavernous ICA. The sections at the aneurysm have become nearly identical in diameter, reaching commonly observed sizes. Although it is not verifiable from imaging, these effects are likely caused by the mechanical forces exerted by the stent and the progressive remodeling of the artery.



	Preoperative		Postoperative		Literature [7]	
	d [mm]	T [%]	d [mm]	T [%]	d [mm]	T [%]
Cavernous ICA	4.16	-	4.14	-	4.13	-
Clinoidal ICA	3.73	10	3.33	20	3.98	4
Proximal	2.84*	32	2.76	33	2.86	31
Distal	2.15	48	2.71	35	2.70	35

Figure 1.3: Vessel diameters (d) and Tapering (T) of the vessels with respect to the cavernous ICA section. Literature values for women were taken from a reference study by Baz et al. [7]. The diameter (*) is the circular equivalent of the area of the elliptic cross-section. Red-tainted values mark that they are outside the standard deviation of the cited work.

1.1.3 Aneurysm Properties

Morphology

Before the operation, the aneurysm enveloped its parent vessel. It had a nearly round shape with a diameter of approximately 25,mm. Post-intervention, the stenosis diminished considerably, and the parent vessel achieved a circular shape. Furthermore, the flow-diverter stent (FDS) retracted the parent vessel from the aneurysm, creating a clearly defined neck instead of the vessel traversing the aneurysm. One week before the rupture, when the patient reported severe headaches, MR images suggested thrombus initiation in the section of the aneurysm facing the MCA.

A week later, the aneurysm ruptured, showing a large thrombosis at the same location. In the rotational DSA images, the aneurysm appeared to have half of its original volume, with missing parts in the suspected thrombosis zone. The rupture

120 occurred at the tip of the aneurysm adjacent to the lateral ventricle⁵, as indicated
121 in Fig. 1.2.

122 Movement

123 The displacement of the aneurysm wall induced by the cardiac pulse became visible
124 only after the rupture. The movement resembled an inflating and deflating bal-
125 loon, with no noticeable displacement at the neck and progressively more significant
126 movement towards the tip. At the tip, the aneurysm bordered the ventricle, a space
127 filled with cerebrospinal fluid that allows movement. Quantitative displacement was
128 difficult to measure due to the rotating video angiography (see the left of Fig. 1.4
for reference).

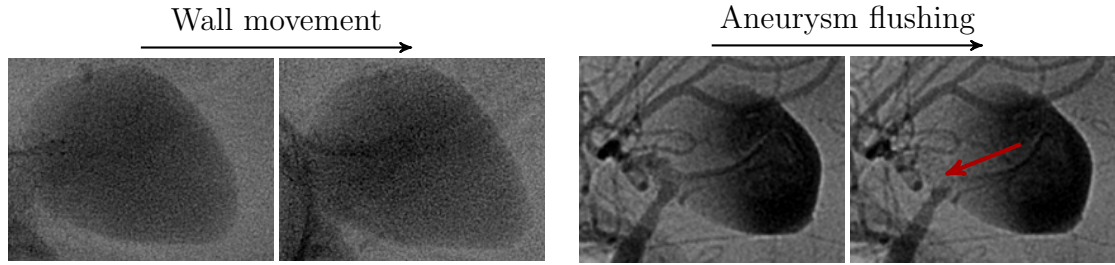


Figure 1.4: An attempt to display the different facets of the movement in the aneurysm. The frames of the contrast agent-filled aneurysm at diastole and systole (left) show distension of the bottom and right faces of the aneurysm. The frames of the aneurysm during ink injection (right) at subsequent snapshots ($\Delta t = 0.17\text{ s}$) display the deflation of the aneurysm, flushing out notable quantities of inkless blood and rendering the MCA transparent (red arrow).

129

130 During the segmentation process, it was observed that the post-rupture images were
131 not as clear as the preoperative ones, particularly due to the absence of contrast dye
132 in some essential vessels. This is evident in Fig. 1.2, where the MCA appears nearly
133 hollow, with only the marker wires indicating the location and size of the vessel.
134 Inspection of the videos taken during injection revealed a remarkable response of
135 the aneurysm to the pressure rise caused by the contrast dye bolus being flushed
136 through the system. The dye is injected following medical protocol at a rate of
137 3 mL/s (60% of the mean flow rate of the ICA) for 7 s. At systole, the dye pene-
138 trates the aneurysm without visible reflux of fluids leaving the bulge. Immediately
139 after systole, the aneurysm deflates flushing out non-saturated blood and rendering
140 the MCA transparent (see the right of Fig. 1.4). Although this effect is only observed

⁵Brain ventricles: A communicating network of cavities filled with cerebrospinal fluid.

during the injection and under increased pressure, it underscores the potential volume variation of the aneurysm and its pulsatility. When the dye injection stops, the reflux of the aneurysm is no longer visible, indicating different fluid dynamics at play.

1.1.4 Simulations

The medical images have revealed important information about the case including topology, morphology, and some dynamics. This information is now enriched by hemodynamic simulations, which will help understand flow structures, pressure evolution, and wall shear stress in the aneurysm. These factors are potentially decisive in the remodeling and rupture dynamics, as discussed in previous publications on similar aneurysms [3, 5, 8].

CFD Setup

For the CFD study, two 3D angiography images are available: one from before the stent deployment and one after the rupture. Both DSA images were segmented following the guidelines described in the previous chapter and meshed with a mesh size of $h = 0.25$ mm. Due to the increased flow velocities at the stenosis, the preoperative case was run implicitly with a timestep of $\Delta t = 0.5$ ms. The convection-related instabilities at peak systole that motivate this strategy were not observed in the stented case, where semi-implicit time stepping was sufficient.

The inflow conditions were imposed using parabolic profiles according to averaged waveforms for individuals in the same age group as the patient [9]. We assumed the inflow conditions to remain the same pre- and postoperatively [8], though we acknowledge this as a potential source of bias, as discussed by Cebal *et al.* [8]. Outlets were modeled using linear resistances as described in Chapter 3. The targeted flow was chosen to depend on the cube of the MCA and ACA diameters, adhering to Murray’s Law. Blood rheology was imposed through the modified Casson model presented in the second chapter. The hematocrit was measured at every visit, resulting in an average of 34.1%, which was used in all computations.

FSI Setup

Since the aneurysm demonstrated clear pulsatility, we investigated the implications of considering rigid arterial walls by comparing our results to fully coupled FSI simulations of the case. These were carried out in cooperation with Aurèle Goetz, another PhD candidate involved in the project. The fluid dynamics are solved as described in previous chapters, with the nuance that an Arbitrary Lagrangian-Eulerian (ALE) description is employed. The mesh is adapted to follow the wall

motions by solving a diffusion equation as done in the work of [10, 11]. Arterial tissue is modeled as hyperelastic using a Neo-Hookean formulation with a Simo-Taylor volumetric model [12, 13]. The solid dynamics equations use a mixed formulation in displacement and pressure and are also solved with a VMS-type method [14]. The coupling between the two solvers is organized in a partitioned way with an under-relaxed fixed-point algorithm [15]. Fluid stresses are applied as Neumann conditions on the solid interface, and solid dynamics yield displacement updates for the fluid. Iterations continue until convergence is achieved, with a tolerance set to 10^{-5} mm on the average solid displacement. An Aitken Δ^2 scheme is used for momentum acceleration, enhancing computational efficiency and convergence [16, 17].

The aneurysm wall is computed as an outward extrusion of the segmented lumen. Due to the contact of the aneurysm membrane with surrounding brain tissue over most of the bulge, a rather high thickness of 0.5mm is employed. The material properties are fitted to obtain movement in agreement with the observed displacements, with greatly diminished stiffness at the dome, where the aneurysm has been observed to pulsate mostly. For reasons of computational feasibility, the FSI case was meshed with a lower spatial and temporal resolution as shown in Table 1.1.

Computation Times

The simulation jobs were executed on 64 cores of the previously mentioned computing facility (see Chapter 3). The computation times with the different integration modalities are shown in Table 1.1. The preoperative case required longer computation times due to the 2–5 subiterations per time step needed for convergence. The postoperative case, computed as a CFD simulation, was faster, requiring approximately half the time per cardiac cycle. The FSI simulation of the ruptured aneurysm took the longest time, at 50 hours per cycle, despite its lower spatial and temporal resolutions. The explicit coupling between the solid and fluid solvers, along with slower convergence rates for the solid solver, are the main reasons for this increase in computation time.

Pre-operative hemodynamics

The aneurysm hemodynamics before the insertion of the FDS are characterized by a high-velocity jet (1.8 m/s) at the stenosis entering and rolling through the formation around a horizontal axis (see the first quadrant of Fig. 1.5 and Fig. 1.6). It then spreads out over the aneurysm surface, generating a large stable vortex in the bulge. As a consequence, high OSI values form on the “eyes” of the vortex, as seen on Fig. 1.5. The WSS in these regions displays persistently expansive properties with high divergences of the normalized WSS vector field. A pouch at the aneurysm’s

Case	h	Time integration	Δt	Comp. time
Preoperative CFD	0.25 mm	Implicit	0.0005 s	34 h
Postoperative CFD	0.25 mm	Semi-implicit	0.0005 s	18.5 h
Postoperative FSI	0.3 mm	Semi-implicit	0.001 s	50 h

Table 1.1: Computation times for the pre and postoperative cases, including the FSI simulation from the PhD work of A. Goetz. Note that the Rigid-FSI comparisons that have been included from the cited work, stem from the same (coarser) fluid mesh.

inner side stands out with high OSI and small TAWSS values, due to its irregular perfusion (see Fig. 1.5).

Post-operative hemodynamics

In the postoperative state, the FDS induces changes of the parent artery, which is now circular and no longer traverses the aneurysm sack. Consequently, the impinging jet at systole is slowed down to half of its preoperative value (0.8 m/s). The volumetric flow traversing the stent at this same instant is 38% of the ICA's flow rate, and the spatio-temporal mean bulge velocity drops to 37 mm/s, as reported in the previous chapter. Notably, this mean velocity increases by 17% when wall compliance is modelled. This underlines the interest of FSI simulations in assessing FDS outcome, as velocity reduction is the ultimate goal of the device. Regarding the flow structure, the weakened inflow no longer holds a stable vortex, but one that travels from near the inflow towards the aneurysm's tip. The OSI experiences an increase on its path, as can be seen on the second quadrant of Fig. 1.5. The flow pattern of the FSI simulation resembles largely the ones from the CFD simulations. The OSI, similarly to reported observations [14], increases moderately when simulating the effect of compliance. Other changes involving flow structure are seen, but not discussed here due to their minor impacts on the discussion. For more details, we refer to the thesis of Aurèle Goetz.

The Pulsatility Index (PI) in the postoperative state was computed at 1.24, resembling common literature values for giant ICA aneurysms [18]. The PI belongs to the rather lower spectrum of values found, which is tied to two main factors. On one side, the impingement angle of the flow and size of the IA permit the formation of large vortices that retain a large proportion of their momentum throughout the cycle. The mean bulge-averaged velocity $\bar{u}$ therefore remains high, and so does also its minimum value. The second factor is the chosen patient waveform. The averaged curve presents two closely followed peaks at systole, instead of one more pronounced

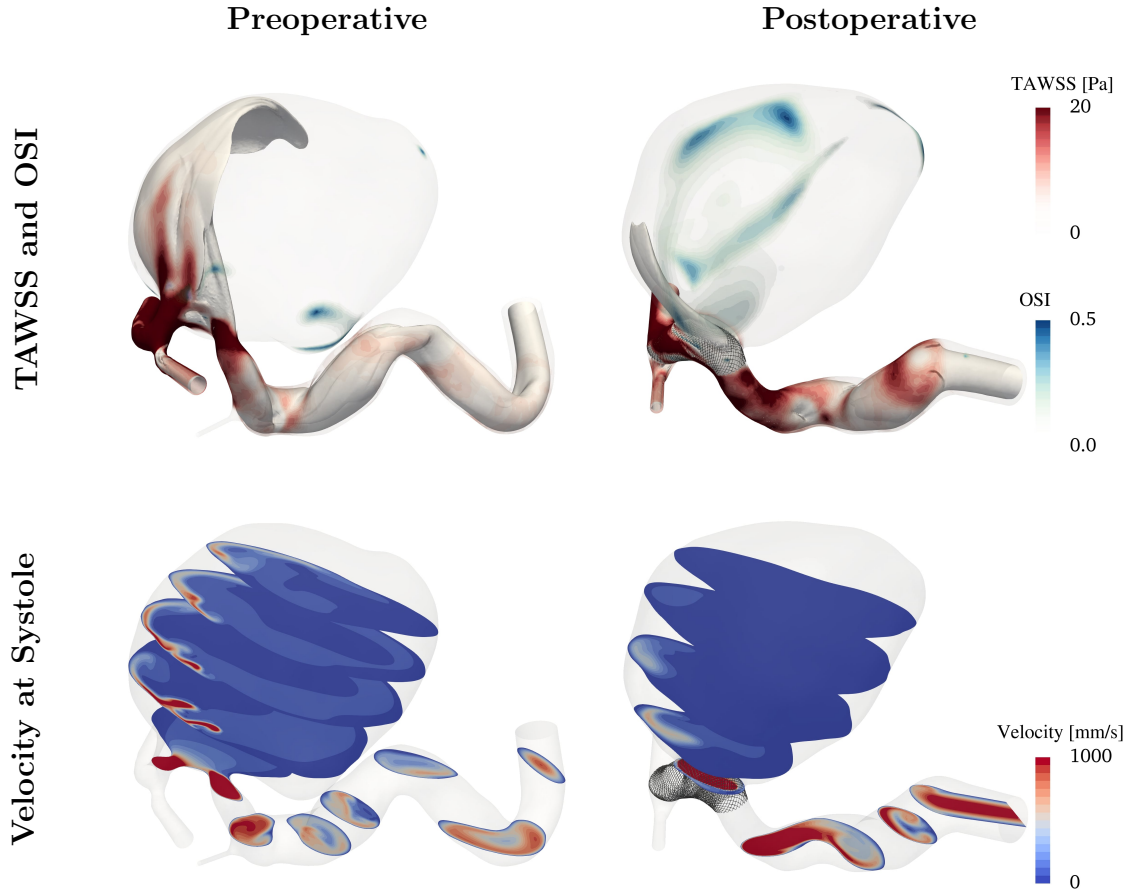


Figure 1.5: Preoperative and postoperative hemodynamics of the aneurysm.

one. This leads to a more regular flow pattern and thus a lower PI.

Pressure changes

Considered separately due to its alleged importance is the alteration of the bulge pressures between the preoperative and postoperative (ruptured) cases. To that end, we have extracted the pressure along the centerline of the ICA in all simulated cases, including the FSI computation. Figure 1.7 shows the pressure drops with a colour-coded inflow volume rate. The pressures have been offset to have zero pressure at diastole in the distal section of the MCAs.

In the CFD simulations, the profiles overall vary almost linearly for the lower flow rates, except for some fluctuations at systole. Notably, the pressure drops in the later ICA sections are remarkably high with respect to the human pressure levels. For comparison, the total pressure drop from the heart to the venous system lies be-

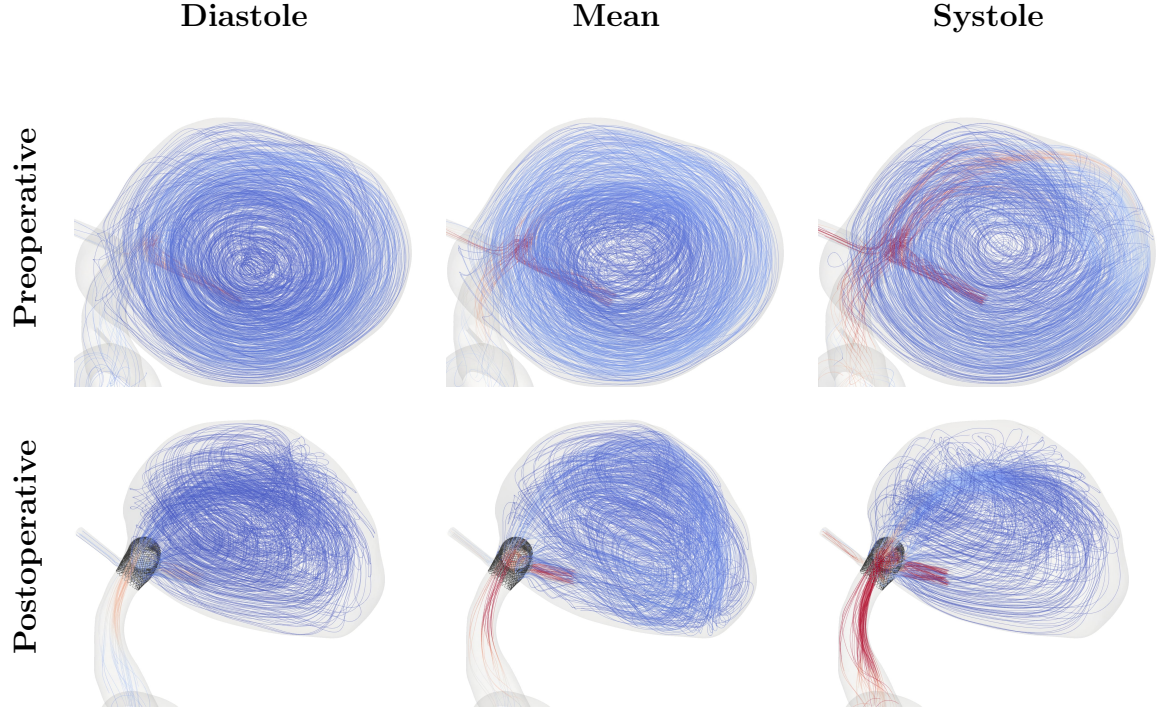


Figure 1.6: Preoperative and postoperative hemodynamics of the aneurysm.

251 tween 80–120 mmHg for a normotensive person, which amounts to 10.7–16 kPa. The
 252 question of whether these are meaningful values or if there is something erroneous
 253 in the modelling remains unaddressed despite being commonly observed [8, 19, 20].
 254 After estimating the pressure drop with linear Poiseuille resistances, we found that
 255 the obtained results indeed are expected. Most of it is the product of the consider-
 256 able tapering that the ICA experiences and possibly the fact of not modelling the
 257 flow through the smaller vessels and perforators⁶. We also confirm through the FSI
 258 simulation, that the large capacitance of the aneurysm leads to a reduction of the
 259 maximal pressure by 1.5 kPa, thanks to a flattening of the pressure waveform over a
 260 cardiac cycle (see Fig. 1.7).

261 A search in the anatomy of the human brain revealed other possible factors that can
 262 be related to the stronger pressure drop in the main arteries. Liebeskind *et al.* [21]
 263 elaborates on the special mechanisms that the brain has to regulate its blood flow.
 264 The brain arterioles lack capillary sphincters that are elsewhere strong pressure
 265 regulators. This shifts the task towards upstream arteries, which need to adapt
 266 seamlessly to changing blood flow conditions, *e.g.* during physical exercise or sleep.
 267 For this purpose, brain arteries have a well-developed layer of smooth muscle cells

⁶Perforator: Small artery penetrating or traversing muscles or bones

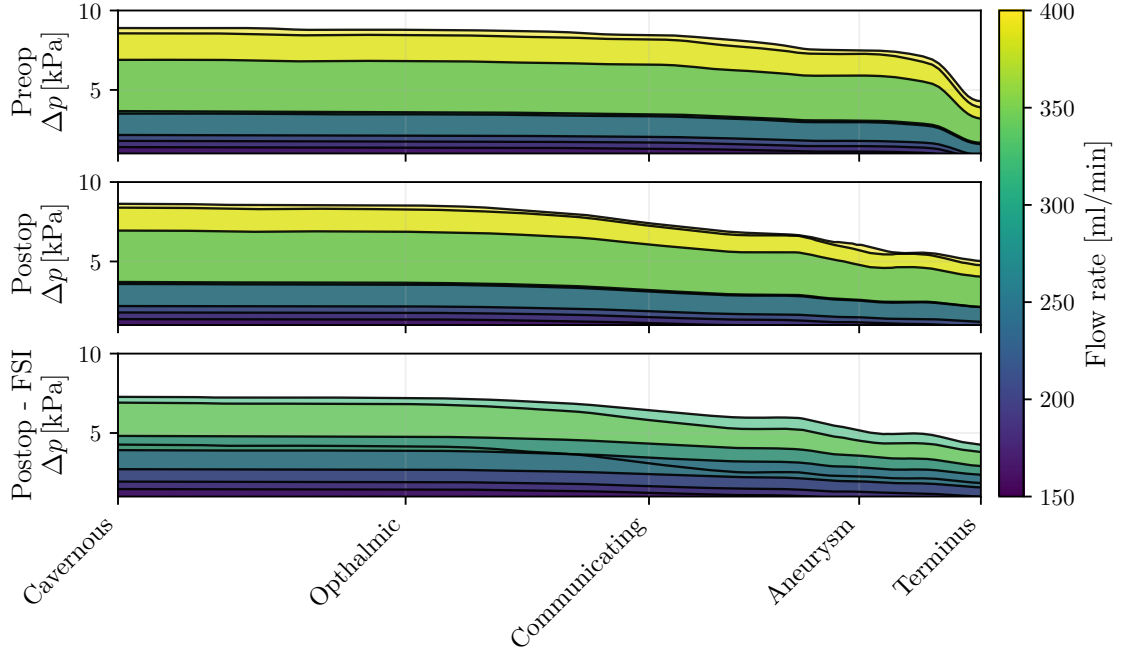


Figure 1.7: Pressures along the centerlines of the ICA and its daughter vessels ACA and MCA in the pre-stented (top) and post-stented (bottom) scenario. The shows the pressure progression in the centerline leading to the MCA whereas the right column does the same for the ACA. Note that here we only display the centerline pressure. It is fairly constant along the vessel cross-section except at the ICA terminus, where the flow split and the inertia can cause differences of up to 200 Pa on one slice.

that can widen or narrow down the vessels, thereby adjusting the flow resistance [21]. The extent of their capabilities is showcased during vasospasm, a critical form of autoregulation that can onset in during haemorrhages. There, the entire ICA may be shut by the medial contraction [22], posing life-threatening conditions. It could thus be possible that the brain arteries have a higher resistance to flow than other arteries, as part of their “standby” autoregulation capabilities. The writer underscores that this is merely a hypothesis and needs to be further investigated.

Returning to the matter at hand, the pressures over the cardiac cycle show mostly a linear relationship with the flow rate, excepting the FSI simulation, where a dephase of pressure and inflow is apparent. For a clearer representation, consider the centerline pressure during systole alone displayed in Fig. 1.8. There, we have equalized the pressures at the inflow according to the downstream autoregulation hypothesis of Cebal *et al.* [8]. In the cavernous sections, pressure drops minimally until halfway into the clinoidal section. There, the tapering of the ICA begins, leading

to a fall of 200 Pa. At the aneurysm, the pre-treatment case sees a flat pressure profile, whereas the post-treatment case continues its nearly linear decay. At the distal end of the aneurysm, the tables turn, causing a sharp drop in the preoperative pressure of nearly 26 mmHg due to the distal stenosis and the sudden contraction of the flow. The postoperative FSI case displays an identical behaviour to its rigid counterpart until reaching the aneurysm. At that point, a surplus of flow enters the bulge causing it to inflate. Consequently, the resistance in the vessel diminishes yielding a pressure plateau.

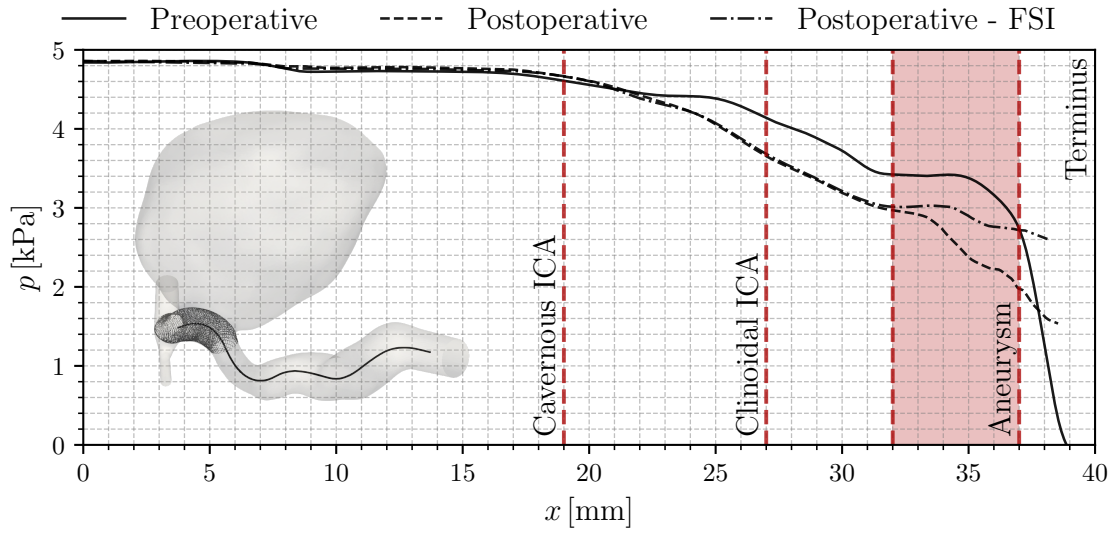


Figure 1.8: Pressures along the ICA centerlines at peak systole. All three pressure curves were offset to have the initial pressure of the preoperative case. Notable drops are seen during the tapering of the ICA and at the stenosis of the preoperative case.

1.1.5 Discussion

With the observed hemodynamics and the patient case description, we can attempt to relate the rupture to the proposed theories from the literature. By doing so, we can at least see if the present case fits into the general picture and if so, support one of the claims with this one case. The proposed theories rely on pressure-induced [8], haemorrhage-induced [4] and lastly flow structure-induced rupture [5].

Pressure induced rupture

In a widely known computational study in the year 2010, Cebal *et al.* [8] analyzed the hemodynamics of 7 FDS-treated intracranial aneurysms located at the upper ICA sections. Of those, 3 ruptured within a week of the operation, while the remaining 4 occluded during the following months. The healed aneurysms were all middle-sized excepting one giant aneurysm, whereas all ruptured belonged to the latter category. In the study, the cases were computed in preoperative and post-rupture states following similar steps as here. The pressures were levelled at the inlets of all simulations to be normotensive. They observed that under those conditions two of the three cases resulted in a pressure increase in the IA. The deployment of the FDS had opened narrow passages of the parent artery located upstream of the lesion, reducing the resistance of the flow and thus carrying higher pressures farther. In the other ruptured case, the stenosis was located past the aneurysm and thus no noticeable alteration of the pressure was reported.

While this hypothesis might be a lead in some cases [3], it seems overall unsuited here. Firstly, the pressure decreases post-treatment due to the location of the stenosis, similar to other works on this subject [5]. In that sense, this aneurysm resembles the giant one that did not rupture in their study. Furthermore, we have observed a strong pulsatility of the aneurysm, which translates into a capacitance that spreads out the pressure waveform. Inevitably, the wall movement introduces further mean pressure losses due to dissipation of energy into the surrounding tissue. Lastly, the timescale of the present rupture exceeds those of Cebal *et al.* [8] by months, pointing towards other long-term remodelling phenomena.

Thrombus mediated rupture

Just months after the previous publication, Kulscar *et al.* [4] proposed a different mechanism for the delayed rupture of aneurysms. In their study, 13 patients with ruptured intracranial aneurysms were studied post-mortem through imaging and autopsies. The patients harboured giant-saccular and fusiform aneurysms that were treated with FDS, and in one case with stent-assisted coiling. 4 out of the 13 patients reported strong headaches prior to the rupture event, similar to our case. The ruptures were split into two categories, early ($n=10$) and late ($n=3$), with the latter taking place at least 3 months after the intervention. The latest instance of a rupture took place after 5 months, which falls short of the 8 months of our case. There, however, double anti-platelet therapy was reduced to single after three months, whereas here the patient was kept on the same medication during the whole time.

One common feature among all cases was the high degree of thrombosis in the

aneurysm. Interestingly, thrombosis is generally considered as the treatment goal of FD and thus a positively perceived event. The two dissected aneurysms displayed a large degree of autolysis⁷ and subsequent necrosis of the tissue. This was expressed in a strong mural thinning and the loss of smooth muscle cells and fibrous tissue. Based on these observations, the authors suggest that thrombosis might have disturbed the balance of the aneurysm wall, leading to a sped-up rupture. The size and aspect ratios (> 1.6) played in thus far a role as that the thrombus forming more quickly within these aneurysms.

In the present giant aneurysm thrombosis is likely to have happened based on the hyperdensity on the MR images, although no autopsy was conducted to confirm it. It was first observed one week before the rupture and grew considerably until rupture. These observations fall in line with the disease progression outline by Kulcsar *et al.* [4]. What remains questionable is whether the thrombus could unchain autolysis to the point of causing a rupture just few weeks after its initiation. In Kulcsar *et al.* [4], they report one particular case in the late rupture group that after 3 months showed full patency of the aneurysm with no other apparent changes. 20 days after switching to single anti-platelet therapy the aneurysm ruptured and was largely thrombosed. If autolysis was the process leading to rupture in that case, it could be possible that little time is needed for a fatal degradation.

Flow structure-mediated rupture

The last proposed theory that will be discussed is presented by Li *et al.* [5]. In their work, 10 patients with delayed aneurysm rupture were compared to 20 control patients, whose aneurysms remained stable. Simulations of the hemodynamics were carried out for both groups, including the simulations of the stents and porosity modelling of coils, when present. A large number of flow indicators, including TAWSS and OSI, were compared. Among the indicators, there was one categorical property related to flow stability. The latter was described as, “flow patterns that persist (do not move or change) during the cardiac cycle.” according to a prior work from Cebal *et al.* [23]. When a stable flow was maintained after stenting, the aneurysms were less likely to have ruptured. The authors proposed that the structure of the flow could likewise impact the thrombus formation. By doing so, their theory could be related to the previously presented thrombus-mediated rupture, although evidence was not provided.

In regards to our case, we can confirm to have transitioned from a “stable” vortex preoperatively to a more irregular flow postoperatively (shown in Fig. 1.6). This transition can also be observed qualitatively in Cebal’s paper [8]. There is however

⁷Autolysis: Self-digestion of tissue.

369 a certain degree of vagueness in the determination of stability that hovers men-
370 acingly over the interpretation. Here, newer flow structure identifiers introduced
371 on the analysis of the topological skeleton [24] could provide a better quantifiable
372 measure of the flow stability. Although first efforts are being made to apply the
373 topological skeleton to intracranial aneurysms[25], clear guidelines for analysis are
374 not yet available.

375 In the discussion regarding “stability”, we have observed the “unstable” flow to be a
376 result of the change of shape of the vasculature and not to result from the embedded
377 stent. Since the work of Li *et al.* [5], seems to consider only the postoperative state,
378 it is difficult to compare the results to our case. Likewise, the timescales of the
379 ruptures are not provided, making a relationship with the present case somewhat
380 impossible. Li *et al.* [5] also includes the modelling of coils as a porous medium
381 in their work. While this is undoubtedly a reasonable strategy to compute mean
382 velocities and other smooth quantities, it remains questionable whether factors like
383 flow stability in the bulge are the same as when embedding the coils into the mesh.
384 From a medical viewpoint, the use of coils did not influence the likelihood of delayed
385 aneurysm rupture, which was also confirmed elsewhere [3, 4].

386 Limitations

387 In the simulation of the case, no patient-specific inflowrate was available. To test
388 the sensitivity, the preoperative case was simulated using flowrates scaled by $\pm 25\%$.
389 Although the flow patterns remained similar, we acknowledge that the nature of
390 the waveform can impact the oscillatory components of the flow and needs to be
391 tested further. Due to the poor quality of the postoperative images at the outlets,
392 segmenting them was difficult and could feed back into the outflow boundary condi-
393 tions. Lastly, it is important to point out that the mesh resolution at the stent was
394 chosen following suggestions from literature [18, 26] instead of convergence studies.
395 This should be included in further work, to safely confirm the parameter choices.
396 Furthermore, it would be interesting to study giant aneurysms using minimally dis-
397 sipative solvers, such as the ones proposed by Valen-Sendstad [27], to assess solver
398 sensitivity.

Conclusions

After conducting a thorough analysis of the flow dynamics of the present giant aneurysm, we contrasted the observations with hypotheses on the rupture pathways found in literature. We have confirmed that under the same modelling assumptions as in the pressure-mediated rupture, no pressure increase could be observed. Thrombus-induced autolysis appears to be the most likely hypothesis, but requires more supporting data to be considered bullet-proof. Methodically, we have seen that FSI can play a crucial role in these cases, where uncertainties around the wall constitution can be overcome through imaging data. The alteration of the pressure dynamics particularly can have a strong impact on the conclusions and should be addressed when observed. Further, we point out some shortcomings of literature in this field, particularly the use of clearly defined indicators for (un-) “stable” flows and also the modelling of thrombus formation, to obtain a better grasp of the temporal progression of the wall weakening. With these careful considerations, researchers might ultimately provide helpful indications to clinicians, who so far, have to rely mostly on intuition when dealing with these intricate cases.

399

1.2 Wide and Blood Blister Aneurysms

A further application to flow diverter stents is the research around blister aneurysms, formally known as Blood Blister-like Aneurysms (BBAs). These formations are uncommon, highly dangerous and poorly understood [28, 29]. In the following section, we will provide a brief introduction to the specific properties of these aneurysms and present a patient case that has been followed for some months at the cooperating medical institution. Additionally, a second ruptured wide-necked aneurysm is included that, although not categorized as a BBA, shares many structural and pathologic traits. With these two cases we once again apply the computation framework to obtain the hemodynamics in pre- and postoperative state. As a hypothetical scenario a second stent is deployed telescopically into the first one to

1.2.1 Pathology

Blister aneurysms are rare but distinct cerebral vascular lesions. They constitute less than 2% of all intracranial aneurysms and are typically found on internal carotid artery, although can also be found in other locations such as the anterior communicating artery or basilar trunk [30, 31]. Blister aneurysms are usually small, shallow, broad-based, or semi-fusiform protrusions on the arterial wall. The aneurysm wall is markedly weak, predisposing these lesions to rapid growth, rerupture, and catas-

418 trophic consequences. They are thought to result from a tear in the arterial inner
419 wall, leading to an intramural haemorrhage and a protrusion covered only by the
420 adventitial layer. Some studies have shown signs of dissection, such as a double lu-
421 men or arterial narrowing. An alternative theory suggests that blister aneurysms are
422 pseudoaneurysms, essentially representing a focal defect in the arterial wall covered
423 by an organized blood clot [31]. The exact aetiology of blister aneurysms remains
424 unclear although high fluid stresses are thought to be implicated in their forma-
425 tion [30, 32, 33].

426 Most blister aneurysms present with subarachnoid haemorrhage (95.6% [30]), diag-
427 nosed initially via brain CT scans. Due to their shape and fragility only a few
428 treatment options are viable in both surgical and endovascular fields. The choice of
429 method is conditioned frequently by the environment and accessibility to the lesion.
430 In absolute terms, endovascular approaches lead in 85% of cases to a functional
431 neurologic outcome compared to 72% from surgical interventions [30]. The authors
432 note however that the success is also conditioned by the severity of the case, which
433 is often more elevated in surgery candidates. An improving trend when transition-
434 ing to endovascular approaches is also reported by others [29, 32, 33], in particular
435 for the recently introduced flow diverters. Independent studies [34, 35] have applied
436 flow diverters to cover the ruptured BBAs obtaining success rates between 90 and
437 97% when using a single device. Using overlapping stents and flow diverters lead
438 to a reduced successrate of only 62%, caused by a higher morbidity [35]. Occlusion
439 rates however were similar.

440 This leaves room for a hypothesis worth studying in an attempt to improve patient
441 treatment. The use of overlapping flow diverters increases the metal cover and the
442 exclusion of the aneurysms from flow. Despite that, the occlusion rates have been
443 observed to be similar. If we assume the increased morbidity to be a consequence of
444 thromboembolic events sparked by a higher quantity of metal in the artery, saving
445 material could decrease the risks for patients without impacting too much the occlu-
446 sion rates [36]. The side-wall nature of BBAs [31] promotes the flow reduction with
447 FDS, since shear driven circulations are very effectively cut by these devices [37].
448 For that reason, we will study the flow in two patient-specific cases on preoperative,
449 single and double stented scenarios. We hope that a description of the hemodynam-
450 ics in these, so far, successfully treated ruptured cases serves as a motivation for the
451 community to conduct larger and statistically meaningful studies on the subject.

1.2.2 Patients' Medical Records

Patient 1

A 56-year old hypertensive female presented to the hospital showing signs of a brain haemorrhage. After an initial MR scan, a blood leak was located in the subarachnoid space close to the terminal left ICA indicating a possible aneurysm rupture. After closer examination the presence of a ruptured BBA was confirmed on the communicating segment of the ICA, upstream of the PCom. The formation has a height of approximately 1.3 mm and expanded over a 4×2.6 mm patch in the axial direction. Prior to the operation, the parent ICA displayed a contraction downstream of the lesion, which is a common reaction to an haemorrhagic event [31] (see Fig. 1.9). After deploying a 4.75×20 mm Silk Vista flow diverter, the narrowing dissappeared. The distal end of the device covered the PCom and terminated before the ACha. Proximally, the stent stretched over the carotid syphon to ensure a proper wall apposition.

After segmenting the postoperative images, the same stent has been embedded virtually between the aneurysm and the PCom. The hypothetic second stent was chosen to be a 4.25×15 mm Silk Vista stent and was deployed over the original with in a distance of two wire radii.

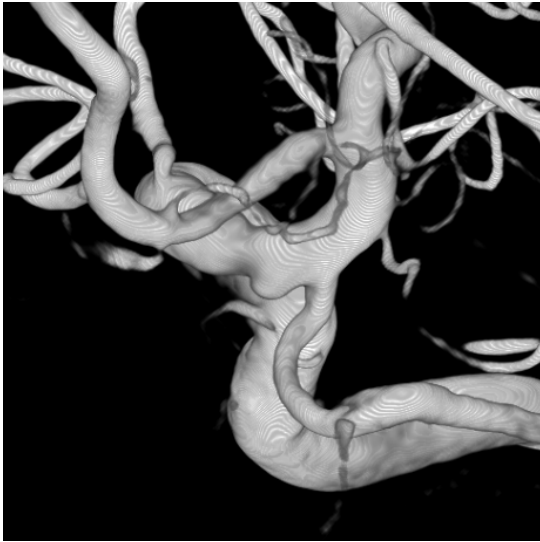


Figure 1.9: Blister aneurysm at the transition between the ophthalmic and communicating segments.



Figure 1.10: Blister-like aneurysm at the proximal end of the cortical middle cerebral artery.

Patient 2

The second patient too presented signs of a SAH and was forwarded immediately to angiographic imaging. The 67-year old male was diagnosed with a ruptures, wide-necked aneurysm at the middle cerebra bifurcation. It measured 1.4 mm at the top and stretched over a larger area of 5.1×2.3 mm as seen in Fig. 1.10. It sits on the first bifurcation of the MCA, between two separating M2 segments, a location known for its complex lesions [38]. Despite the complications of reaching this distal location, the interventionalists managed to successfully deploy a 3.5×20 mm Silk Vista stent across the larger M2 and M1 segments. An added difficulty in this case is the balance between flow redirection and risk of thromboembolic events downstream of the jailed bifurcating artery. Due to the weakness of the arterial wall and the shape of the formation, no coils could be used, much like in an BBA. To evaluate the alterations caused by a second virtual stent we simulated a 3.5×11 mm XFRED device inserted into the previous FDS.

General simulation setup

The geometries were meshed following the same pipeline as described before in chapter 4. The smaller geometries permitted a more resolved isotropic mesh size of 0.15–0.2 mm in both cases, while stent meshing parameters remained unchanged. Simulations were carried out using semi-implicit time stepping with $\Delta t = 1$ ms for 3 cardiac cycles. Boundary conditions and viscosity models were applied in the same manner as the previous case. The in the case of patient 2, the inflow was set at the beginning of the MCA with 50% of the ICA’s flowrate [39] for computational efficiency. The choice is justified due to the length of the M1 segment and the lack of tapering on its way, although the waveform shape may not be necessarily suited.

The obtained computation times per cardiac cycle are given in Table 1.2. Note that number of cores used was conditioned by server availability and not necessarily by optimized distributions.

1.2.3 Simulations

Patient 1

In the preoperative scenario, the flow in the core of the ICA causes a “lid-driven cavity” kind of flow in the BBA. Flow velocities at systole in the bump are close to 0.2 m/s, while pressure gradients between the aneurysm are nearly 0. When executing the virtual stenting for the first device, one obtains an estimated metal coverage ratio of 17% at the aneurysm neck with wire angles at 40° with respect to the flow direction. As seen in the upper row of Fig. 1.12, the recirculation speed

1 Patient-specific analysis with flow-diverter treatment

	Case	Unstented	Single Stent	Double Stent
Patient 1	Mesh Size	1.8M	7.2M	10M
	Computation Time	1:05h	7:20h	7:05h
	# of processors	128	128	196
Patient 2	Mesh Size	2.3M	6.2M	12M
	Computation Time	4:30h	6:15h	15h
	# of processors	64	128	128

Table 1.2: Computation times per cardiac cycle and used resources. Note that number of cores used was conditioned by server availability and not necessarily by optimized distributions. All meshes have an isotropic meshsize of 0.2 mm and a resolution at the stent as described in chapter 4.

	Case	Unstented	Single Stent	Double Stent
Patient 1	mean(TAWSS)	2.0	1.1	0.6
	mean(OSI)	0.010	0.015	0.023
Patient 2	mean(TAWSS)	4.1	1.6	0.6
	mean(OSI)	0.002	0.001	0.002

Table 1.3: Bulge averaged TAWSS and OSI values for both patients in all computed configurations.

at systole is halved after deployment to 0.1 m/s. The flow in the adjacent PCom is altered by the wires, forcing it to enter more perpendicularly now. This feeds back into the flow of the parent artery, as seen in Fig. 1.12. The second fictive stent yields a similar metal coverage of 20% due to its reduced oversizing. Together with the previous device, this results in a joint metal cover of 34%. The value is obtained as a product of the porosities of the devices. The increased resistance reflects upon the reduced flow at the dome tip, whose peak values do not surpass 0.06 m/s.

The shear stresses on Fig. 1.12 shows a reduction and a smoothing WSS values in the stented cases. The TAWSS resembles the systolic WSS with 1/3 of the magnitude, for which we decided to omit another plot. Mean TAWSS and OSI values are listed in Table 1.3 for reference. Note that for the computation of these values, the aneurysms were extracted from their environment at the contact line with the stent. Contrary to the TAWSS magnitudes, which fall for decreasing porosities, the mean OSI tends to grow moderately, due to the slower recirculation.

519 Patient 2

520 The T-shaped bifurcation on which the aneurysm is found leads to an inertia driven
521 circulation at the dome. The pressure on a cutting plane of the case Fig. 1.11
522 reveals the impact zone of the impinging jet at the left end of the aneurysm. The
523 WSS in that area presents large gradients due to the separation of the flow in the
524 two directions. The bulge in the preoperative state has an inhomogeneous WSS
525 profile with elevated values on its tip.

526 The insertion of the first stent yields an estimated metal coverage of 21% and affects
527 the pressure, but most notably the WSS. The latter is reduced and stratified post-
528 operatively. As can be seen on the stent interface, a portion of the high pressure
529 values rests on the stent close to the impingement area. These effects become more
530 notable after deploying the second telescopic stent, as shown in the third column of
531 Fig. 1.11. The new stent, contrary to the Silk Vista models, counts 64 wires and
532 features an elevated metal cover of 29% in its deployed state. When both porosities
533 are factorized, a total coverage of 44% is obtained. The bulge-averaged TAWSS
534 drops as a consequence more strongly than for patient 1 as can be seen on Table 1.3.

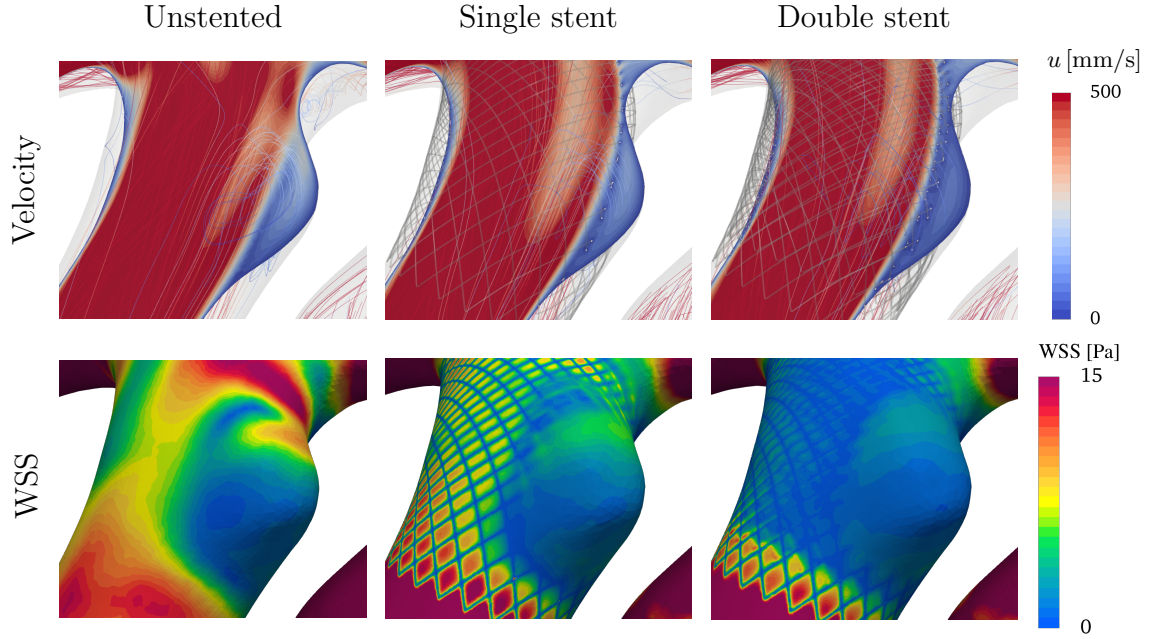


Figure 1.11: Systolic velocities (upper row) and WSS (lower row) for the three computed cases of patient 1. TAWSS and WSS distributions only differ quantitatively which is reflected as well on low OSI values.

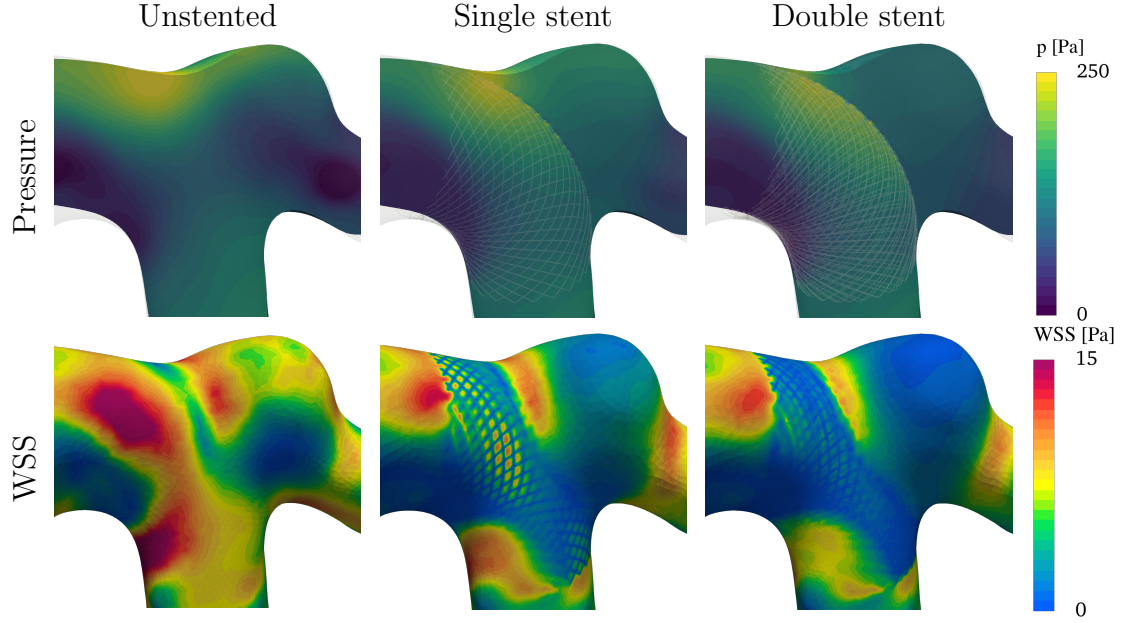


Figure 1.12: Systolic pressures (upper row) and WSS (lower row) for the three computed cases of patient 2. The stent redirects part of the flow, causing a mild pressure drop in the bulge. Pressure gradients in the jailed M_4 are smoothed through each deployed stent.

1.2.4 Discussion

The two presented patients are characterized by complex aneurysms with little margin of freedom during treatment. Their weak walls and shallow morphology restrict clinicians to use only a small subset of the methods that are usually available for handling saccular aneurysms. Although the formations differ in nature, *i.e.* the first is classified as a BBA and the second as a regular IA, both raise the question on how much metal coverage is justified to reduce the hemodynamic stress. While a reduced stress is desirable on ruptured tissue to foster healing and reduce risk of rerupture, an excessive amount can cause serious thromboembolic complications in jailed vessels. While we can certainly not answer this question with only these two patients, we can at least the hemodynamics that, so far, have kept the patients in good shape without neurological and further haemorrhages.

Stenting blister aneurysms

The location and shape of BBA aneurysms makes them commonly subject to unidirectional, high WSS [29]. In our study, we have confirmed a consistent reduction of the flow in the BBA by one-half despite a low metal coverage compared to the

recommended 30%. This result, as already noted by Augsburger *et al.* [37], stems from the good capabilities of FDS to cut down shear-driven flows. If a WSS reduction is assumed to be one of the determinant factors in the healing process of a rupture, the use of a single stent might be preferred. The second FDS, although once more reducing the TAWSS, achieved only a moderate absolute reduction with respect to the first stent, while nearly doubling the metal surface exposed to blood flow. The problematic feature of metallic surfaces is that they are thromboembogenic [40], *i.e.* they can easily trigger coagulation cascades forming blood clots on their surface. These can detach once a certain size is reached and travel down the arterial system causing infarction of vessels. Despite recent efforts in coating the devices [41], (dual) antiplatelet therapy remains mandatory and often prescribed for a lifetime in patients that have an implanted stent [2].

To more accurately assess this research question, it would be necessary to have an estimation of how much exposed metal bears a noteworthy risk. We have seen several studies assessing the thrombogenicity of FDS, but our trust remains low since, all studies are financed by or supported by FD manufacturers and claim at the same time that their newly developed model is better than its competitors [41, 42].

Stenting bifurcation aneurysms

Bifurcations belong to locations where the use of coils or WEB devices is preferred [2]. When these cannot be used, due to a wide-neck formation for instance, flow diverters have to be deployed through the main artery and one of the outgoing branches. Selecting the outgoing branch is conditioned by the size of the vessels, their role in brain perfusion and how difficult the operation is. In this case, the largest outflow was chosen for two reasons, (i) it was better aligned with the parent vessel and (ii) it was more readily accessible with the catheter, permitting a safer intervention. Had the other branch been chosen, then there would be a non-negligible risk of hitting the ruptured aneurysm with the guide device. Additionally, the second M2 had a 48% smaller radius while the chosen M2 preserved the size of the parent vessel (see Fig. 1.10).

Hemodynamically, we confirm that the insertion of the FDS has helped deflecting the impinging flow away from the bulge reducing hemodynamic stress. We also executed the deployment in the other configuration. Due to the notoriously smaller vessel caliber of the second branch, the porosity of the FDS was notoriously high and only adjusted past the aneurysm neck to nominal values. Under those circumstances, the diverting effect would not have had the same impact and furthermore, we expected a redirection into the aneurysm, making it straight ahead not recommendable. For these reasons, we did not pursue this approach any further.

The use of a second stent in this case shares the same criticism as in the previous patient. The chosen stent in this case was an XFRED device which yields a lower porosity than the Silk Vista models. This device also exerts a larger radial force that can, as was the case in the giant aneurysm, change the vessel geometry [43]. The higher metal cover counterbalances the inertial flow and manages to yield similar stress reductions as in the BBA. Nevertheless, it is unclear how the difference in performance ponders with the increased risk of thromboembolism.

Limitations

Similarly to the previous study, we point out that inflow conditions were in this case generalized and can impact the results. In the case of patient 1, we modelled the PCom as a regular outflow, while a well-developed CoW was seen on MR scans. A better choice in this case would be conducting a study on the CoW to extract local flow rates and impose them as boundary conditions, as discussed in chapter 3. In both cases, we utilized the postoperative images for segmentation. In the preoperative ones, there were more perforators clearly visible (ophthalmic and anterior choroidal arteries) that could be added or modelled later for increased fidelity.

Conclusions

Uncoilable aneurysms, such as BBA or wide-necked aneurysms, can benefit from an FDS treatment as is the case for the presented patients and as reported in literature. The hemodynamic alterations of a single FDS have proven sufficient in treated patients until this day. The hypothetical use of a second telescopic FDS has provided limited usage with respect to the first device, making it probably unjustified in the face of the added thromboembolism risks.

In this study, hemodynamics have proven useful in characterizing flow and measuring the impact of the devices. The deployment itself was a fundamental contributor, since it provided the porosity estimates that cannot otherwise be extracted from the deployed devices. With larger studies on cases like these, we hope that clear guidelines in treatment can be defined, helping clinicians in choosing the most favorable treatment for their patients.

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Chapter 2

Conclusions

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2.1 Conclusion

Decades of research in biomechanics have laid a foundation in clinical practice by developing virtual analysis tools for various cardiovascular diseases. These tools have not only enhanced our understanding of the origins of these pathologies but also facilitated the design of optimized treatment strategies. Building on this foundation, both this thesis and the ERC project CURE aim to improve diagnosis and treatment options for patients with intracranial aneurysms. This work focuses on three key research areas: (1) the study of stabilized finite elements for cardiovascular flows, (2) the assessment of the vascular extension around an aneurysm required for accurate numerical simulations, and (3) the development of a fast virtual stenting algorithm for deploying flow diverter stents. Although distinct, these aspects are united by a shared goal of advancing state-of-the-art modeling, with a strong emphasis on the numerical aspects of simulations.

In the second chapter, we delve into the physics and numerical methods for simulating hemodynamics in brain arterial flows. We revisited the existing numerical framework in a didactic manner, identifying an inconsistency in one modeling choice, which led to the research question: “How can a tensor-based stabilization parameter become impervious to the ordering of elements?” To address this, we developed a novel strategy based on Principal Component Analysis, testing it against common 2D and 3D examples from the literature. While not exhaustive, the validation showed improvements over an established alternative and consistently aligned well with experimental data.

With an improved understanding of the FEM backend, we explored the complex technical aspects of brain arterial flow, which is constrained by medical knowledge and limited measurement data. We reviewed the literature on boundary conditions and detailed the segmentation and meshing workflows implemented. This led to another research question: “What is the extent of the vasculature that needs to be simulated around the region of interest to obtain physiological results, particularly when using common outflow models?” This question arises from the segmentation of the anastomotic (redundantly connected) vessel network in the brain, where arterial loops can theoretically direct flow in any direction and do not always follow common splitting laws. We studied the sensitivity of three patient-specific cases with aneurysms, comparing full Circle of Willis simulations with two reduced vasculature models commonly seen in the literature. The results revealed case-dependent variability, suggesting the need for preliminary studies using different imaging sources to avoid modeling pitfalls.

The next focus was on flow diverter stents, which enable clinicians to treat otherwise inoperable intracranial aneurysms. A numerical representation of these devices not

only replicates altered flow conditions but also provides insights into how hemodynamic changes affect vessel walls. We proposed a novel virtual stenting algorithm based on geometric analysis, validated it against literature cases, in-vitro experiments, and postoperative patient images. We then demonstrated how to embed the device into an arbitrary unstructured computational mesh and ran a test simulation. The proposed methodology offers three advantages: (i) accurate estimation of local stent porosities, (ii) robust and fast deployment, and (iii) reliance only on centerline information and manufacturer data for deployment. However, the method does not account for mechanical interactions with vessel walls and may underperform with significant stent undersizing.

Equipped with the essential tools for post-treatment aneurysm simulations, we examined three patient cases involving flow diverter stents with varying outcomes. The first case, featuring a giant aneurysm, suffered a delayed rupture after treatment. Simulations of preoperative and postoperative states including wall compliance were carried out and served an analysis based on several rupture hypotheses from the literature. The remaining cases involved (i) a ruptured blister aneurysm and (ii) a ruptured wide-necked aneurysm. Although different, both cases required assessing similar risks, including the trade-offs between reducing flow stress and managing thromboembolic complications. We report on the computed hemodynamics and include a study on deploying a second stent over the existing one. With these results, we hope to contribute to larger population studies aimed at optimizing treatment options.

2.2 Perspectives

The work is one of the two initial thesis conducted at the start of the aforementioned ERC project CURE. The project posed many initial hurdles thanks to which we learned and progressed. With the momentum gained, the subsequent work of other PhD students can tackle more specific problems. Among all the possible modalities to follow, there are some that are of particular interest in the context of flow diverters.

Firstly, thrombosis has been observed in literature and in one of the presented cases, to lead in some cases to wall autolysis and aneurysm rupture. Simulating thrombosis can help confirming or disproving the autolysis hypothesis by providing the growth dynamics and the exposure time of damaged tissue. Furthermore, it could also be applied to optimize stent design in order to achieve an optimal occlusion rate. The latter optimization has already been conducted by the team, exploiting the capabilities of Reinforcement Learning, but using a reduced simulation complexity [1]. These ideas can be pursued with more elaborate modelling techniques for which

680 parts of the study will serve. The porosity maps from the stenting algorithm can
681 be used in combination with novel continuum models [2] of flow diverters to yield
682 a light-weight, yet accurate method for optimization purposes. The Reinforcement
683 Learning has also been advanced in our team ever since [1] thanks to an improved
684 environment based on partial differential equations [3]. With these tools it may be-
685 come part of clinical routine to design and process personalized medical devices that
686 adjust to each patient's needs at feasible cost.

687 Validation to this ambitious goal is more increasingly available thanks to improved
688 fields strengths and frame rates of Magnetic Resonance and Computed Tomography
689 machines. With the help of Machine Learning in denoising and gap-filling mea-
690 surements, the adquisition of reliable data could be carried out as part of clinical
691 routine. This in turn helps validating new algorithms and tackles at the same time
692 the limitiations imposed by using generic boundary conditions.

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RÉSUMÉ

Sed tamen haec cum ita tutius observentur, quidam vigore artuum inminuto rogati ad nuptias ubi aurum dextris manibus cavatis offertur, inpigre vel usque Spoletium pergunt. haec nobilium sunt instituta.

MOTS CLÉS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method

ABSTRACT

Ego vero sic intellego, Patres conscripti, nos hoc tempore in provinciis decernendis perpetuae pacis habere oportere rationem. Nam quis hoc non sentit omnia alia esse nobis vacua ab omni periculo atque etiam suspicione belli?

KEYWORDS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method